Curriculum Vitae/Resume | Advika Lakshman | Incoming MS DS Student at NYU

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EDUCATION

Shiv Nadar University, Chennai, Tamil Nadu, India 2022– 2026

B.Tech in Artificial Intelligence and Data Science **CGPA: 9/10**

Relevant Subjects: Artificial Intelligence, Machine Learning, Deep Learning, Data Mining, Big Data Analytics, Natural Language Processing, Web Technologies, Image and Video Processing, and Reinforcement Learning

ACADEMIC PROJECTS

Title: Medical Image Inpainting for Brain MRI Nov 2025

- Implemented and compared diffusion-based models (DDPM baseline vs. Diffusion KAN with U-KAN backbone) to restore healthy tissue in tumor-masked T1 MRI slices.
- Achieved better PSNR (20.06), SSIM (0.80), and MSE (0.012) via KAN layers for nonlinear anatomical modeling and RePaint denoising.

Title: Early Sepsis Prediction Jun 2025

- Built a machine learning model to predict sepsis at early stages, using ICU data from the 2019 PhysioNet Challenge.
- Handled missing values in time-series clinical data using linear interpolation and MICE imputation methods; addressed class imbalance and trained models like logistic regression and XGBoost.
- Focused on maximizing recall to reduce false negatives, reaching up to 80% recall with some trade-off in accuracy.

Title: Sketch-to-Face Image Translation using DCGAN Jun 2025

- Created a deep learning pipeline using DCGAN to convert face sketches into realistic images using the CUHK dataset.
- Trained 2 model versions (with Binary Cross Entropy loss & Mean Squared Error), comparing their image quality (SSIM: 0.51 vs 0.55)
- Delivered a clean, modular codebase with loss visualizations, pre-trained model checkpoints, and organized result outputs.

Title: Disentangled VAE-based Intelligibility Assessment for Disordered Speech May 2025

- Designed a multimodal disentangled β-VAE architecture with audio (STFT spectrograms) and BERT text embeddings on the ICASSP 2025 Alzheimer's detection dataset, targeting 3-class classification of healthy, MCI, and dementia speech.
- Implemented fragmentation, augmentation, and fusion pipelines, demonstrating that the disentangled latent space with triplet and total correlation losses yields linearly separable clusters and outperforms baselines and other deep models in validation accuracy and interpretability.
- Manuscript titled "Disentangled Variational Autoencoder based Intelligibility Assessment System for Disordered Speech: A Multimodal Representation Approach" received by the International Journal of Pattern Recognition and Artificial Intelligence and currently under review.

Title: Geophysical Data Enhancement through Masked Autoencoders Mar 2024

- Utilized a suite of Masked Autoencoder models–GAN-enhanced and Vision Transformer-integrated variants–to perform inpainting tasks on seismic data.
- Achieved detailed restoration of geophysical images, aiding accurate subsurface interpretation and resource exploration analysis.

PUBLICATIONS / MANUSCRIPTS ACCEPTED

- **Advika Lakshman**, Prathamesh Devadiga**. "Resistant to Lawyers, Defeated by Disagreement: Evaluation Blindspots in Legal Language Models."** Paper accepted in ICML AI4Law conference workshop held at Seoul July 10th 2026.
- Sayak Dhar, Lavanya Nemani, Shekhar Nath, Keshav Agarwal, **Advika Lakshman**, Shardul Kamble, Swaraj Basu, Prakash Hiremath, Nihesh Rathod, Stan Iyer, Radhakrishna Bettadapura, Ramesh Hariharan, and Badri Padhukasahasram. **"A powerful multi-omic AI framework for Blood-Based Cancer Screening and Tissue-of-Origin Prediction."** Abstract submitted to ISMB 2026, TransMed: Translational Medical Informatics track, Submission #1073. Accepted.
- **Advika Lakshman**, Jeanette Shutay, "**Comprehensive Crop Yield Forecasting in India: A Multi-Model Machine Learning Approach with Population Density Integration for Agricultural Planning"** . Paper accepted at *Exploratio Winter Cohort (Dec 2025).*

PUBLICATIONS / MANUSCRIPTS UNDER REVIEW

- **Advika Lakshman** et al. **"Disentangled Variational Autoencoder based Intelligibility Assessment System for Disordered Speech: A Multimodal Representation Approach."** Manuscript received by the International Journal of Pattern Recognition and Artificial Intelligence. Under review.

INTERNSHIPS

Data Science Intern - Strand Life Sciences Jan 2026 - Present

- Built hierarchical and multiclass machine learning pipelines for CancerSpot, a multi-omic AI framework for blood-based cancer screening and tissue-of-origin prediction across multiple cancer types.
- Developed L1 cancer detection and L2 tissue-of-origin modeling workflows, including sex-specific model routing, threshold calibration, and top-3 tissue-of-origin prediction using model score deltas.
- Contributed to the submitted ISMB 2026 TransMed abstract "A powerful multi-omic AI framework for Blood-Based Cancer Screening and Tissue-of-Origin Prediction" (Submission #1073; under review).
- Worked on DTI improvement and an agentic biomedical database chatbot enabling natural-language querying, grounded responses, and automated analysis across biomedical and clinical datasets.

Research Intern –University of Chicago May 2025 – Jul 2025

- Conducted research on crop yield forecasting by integrating demographic factors (population density, urban–rural ratios, market accessibility) and agronomic features with machine learning algorithms.
- Used Random Forest, XGBoost, LightGBM, & CatBoost, recording best performance with Random Forest ($R^2 = 0.98$, RMSE = 125.79).
- Designed a robust data preprocessing pipeline, handling 8% missing values with imputation and addressing outliers for 19,689 agricultural records across 28 Indian states.
- Published as a research paper under the title "**Comprehensive Crop Yield Forecasting in India: A Multi-Model Machine Learning Approach with Population Density Integration for Agricultural Planning"** in the journal *Exploratio Winter Cohort (Dec 2025).*

Global Academic Intern–Big Data Analytics, Deep Learning, Machine Learning with Generative AI, NUS Jun 2025

- The programs imparted in-depth knowledge of Big Data Analytics & Deep Learning—discussing core big-data concepts, machine-learning algorithms and tooling, and end-to-end pipeline development.
- This further exposed me to Applied Machine Learning with Generative AI—primarily GAN architectures, transformer-based large language models, and prompt engineering & fine-tuning in a research-driven, collaborative environment.

Research Intern-Spring Lab (Speech Technology), Indian Institute of Technology Madras May 2024 – Jun 2024

- **Developed and Implemented ASR Pipelines:** Worked extensively with HuBERT models and ESPnet (End-to-End Speech Processing Toolkit) for building and training Automatic Speech Recognition (ASR) systems, using datasets like LibriSpeech, Spring Speech, and various other Indian language datasets. Successfully debugged and trained ASR pipelines for multilingual speech data.
- **Feature Extraction and K-Means Clustering:** Performed feature extraction and built K-Means models using Fairseq (sequence modeling toolkit) for Hindi, Tamil, and Indian English datasets. Automated units collection tasks, created token lists, and implemented preprocessing scripts to enhance ASR model performance.

Intern, Marketing and Public Relations Department, Shiv Nadar University Chennai Oct 2022-Dec 2023

- Partook in organizing events—contributing to logistical planning, promotional efforts, and ensuring their seamless execution.
- Conducted content writing initiatives to enhance brand messaging and communication strategies.

EXTRACURRICULAR ACTIVITIES AND ACHIEVEMENTS

- Received an Academic Excellence Award for securing Second Rank across the entire batch in the Vth semester.
- Awarded with a merit scholarship by Shiv Nadar University for securing Rank 1 in the B.Tech Entrance Examination (2023).
- Professional Bharatanatyam Dancer with formal training of 12 years at Nrityakshethra Dance School in Bengaluru—Rank II in districtwide Nrithya Vrinda Classical Dance Competition (2023), Rank I in the state-level Youth Power Dance Competition (2023), Rank I in ANASS India Dance Contest Season 5 (2021), and Rank III in ANASS International Online Dance Contest (2021).
- Awarded with a B Grade in Bharatanatyam by Doordarshan, India's public service broadcaster (2024).

SKILLS

ICT: Python, PyTorch, TensorFlow, Pandas, NumPy, Matplotlib, Seaborn, SQL, Tableau, Anaconda, Lightning AI, and MATLAB
Certifications: Introduction to Large Language Models (LLMs) (NPTEL, 2025) | Project Management Fundamentals (IBM SkillsBuild, 2023) | Programming in Python (TechSparx, 2021)

Bangalore - 560048, Karnataka, India